Dilations

1

a No

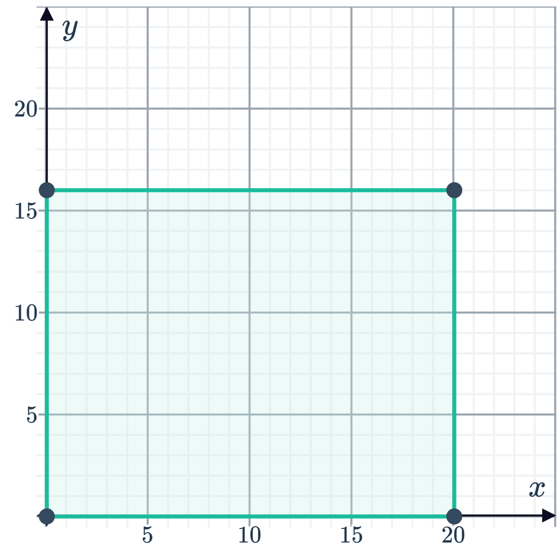
b Yes
Scale factor $= 2$

c Yes
Scale factor $= \frac{1}{4}$

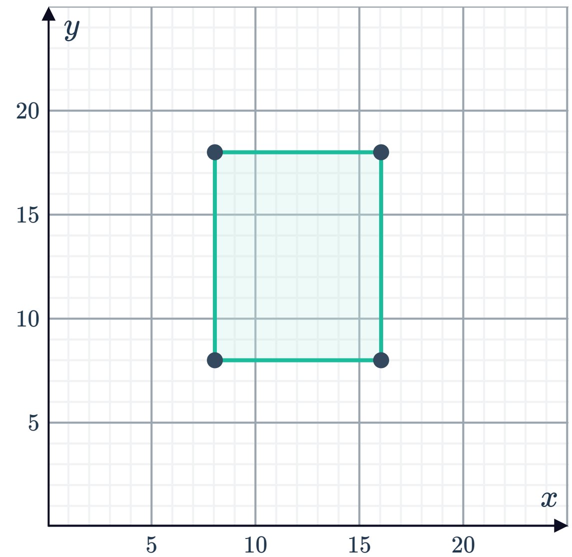
d Yes
Scale factor $= 2$

2

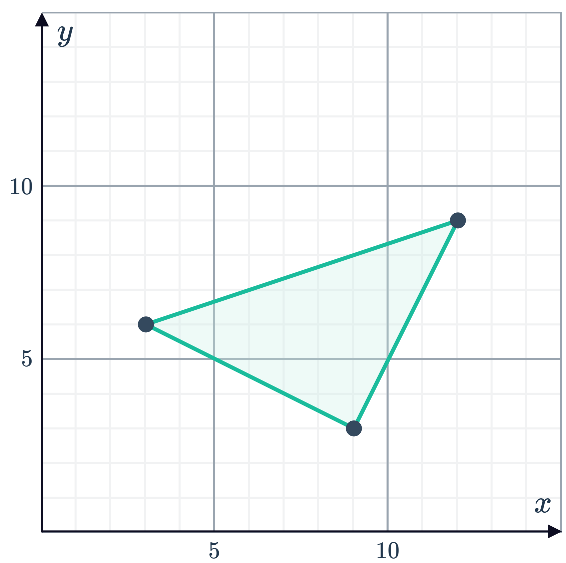
a



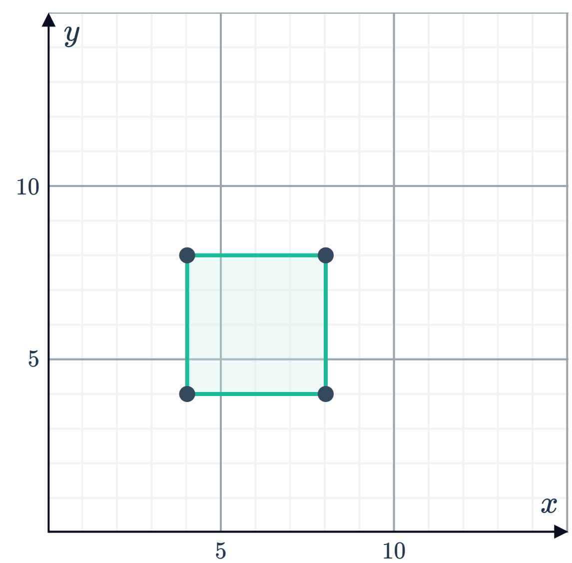
b



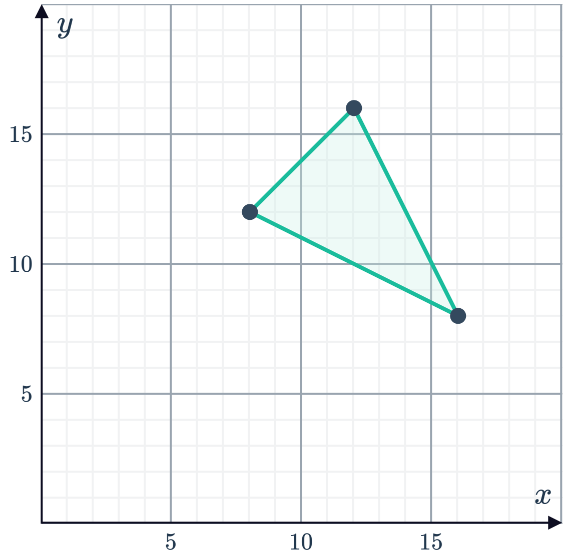
c

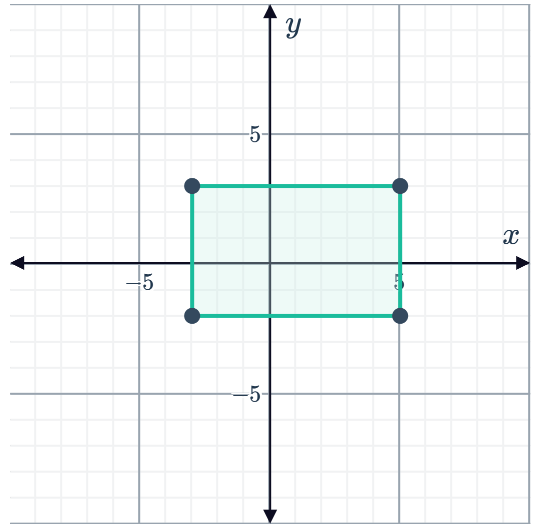


d

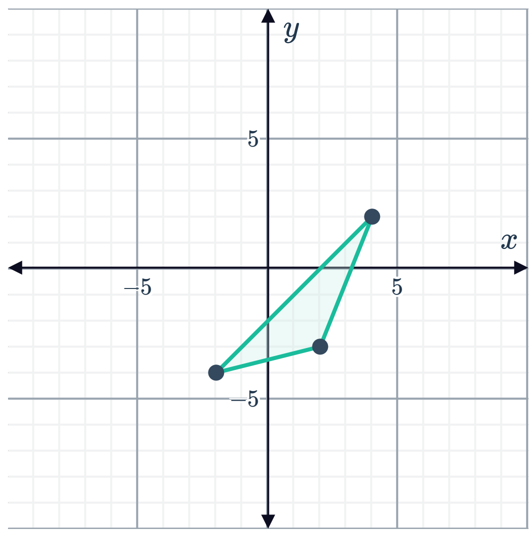


e



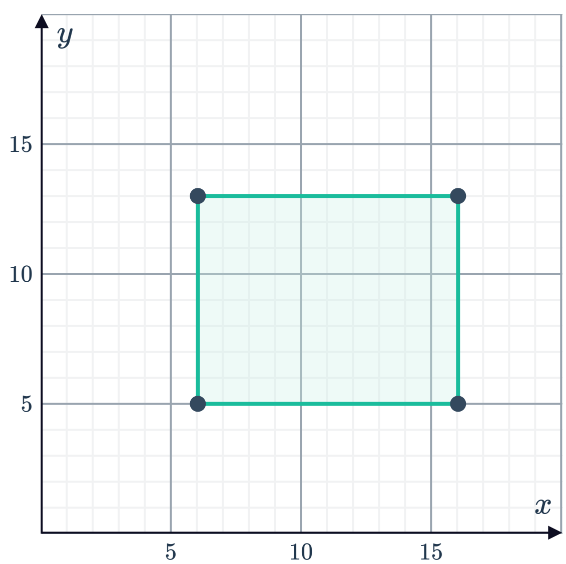


g

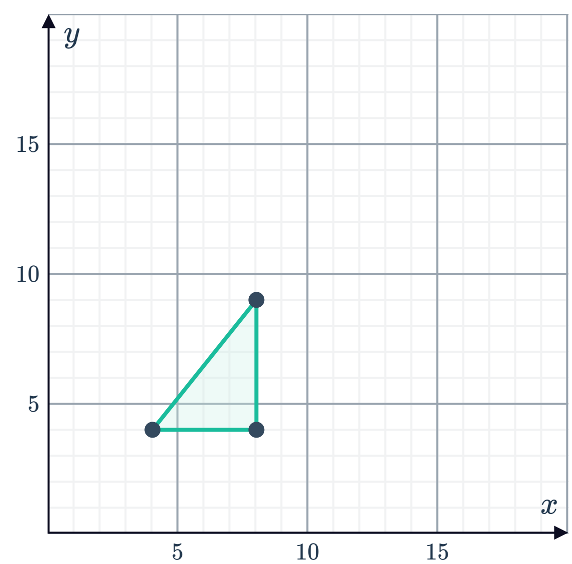


3

a



b



4 The new coordinates will be $A'(-3.5, -2.5)$, [illegible]$5, -2.5)$, and $C'(-2.5, 4.5)$.

5 2 ft

6

a 5

b $\frac{2}{3}$

7 $\frac{1}{3}$

Additional questions

8

a Scale Factor$= \frac{1}{2}$

b Scale Factor $= 3$

c Scale Factor $= \frac{4}{5}$

d Scale Factor $= 6$

9

a False. Scale factors cannot be negative.

b True. Multiplying each length by a factor greater than 1 results in the image being larger than the preimage.

c True. Dilations preserve angle measure.

10 The numerator does not have to be 1. Any fraction in the form $\frac{a}{b}$ where $a < b$ will produce a reduction.

11 Answers may vary.
All correct solutions must indicate that the center of dilation is the center of the circle. Start with a circle with a radius of r. For the first ring, dilate the circle using the center of the circle as the center of dilation with a scale factor of 2, for the second ring dilate the circle using the center as the center of dilation by 3, and for the outer ring dilate the circle using the center as the center of dilation by 4.

12

a Yousuf should dilate his original notes by a scale factor of $\frac{6}{17}$ to fit exactly on the notecard

b Yousef's paper will be 3×3.88 inches after applying the scale factor.

c Answers may vary.
For example, Yousef could divide his sheet of notes in half horizontally making it an 8.5×5.5 inch sheet of paper and then dilating each half by $\frac{6}{11}$ which gives each half of his notes the dimensions 3×4.64inches and utilizes more space on the notecard.